Week :8 (19-23 October 2026)

من - إلى6الأسبوع #:

Subject/الموضوع: Physics **Unit Title/الوحدة عنوان: Projectile motion** **Grade/: 11**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 **قانون بويل**	I can describe projectile motion by treating the **horizontal and vertical components of motion independently**.	2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 2 **الحصة الثانية**	I can apply kinematic equations to **analyze and solve projectile-motion problems**, including projectiles launched at an angle..	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 3 **الحصة الثالثة**	I can identify and represent the **forces acting on an object** and use Newton's laws to explain and predict its motion.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	please check student Teams
Lesson 4 **الحصة**	I can explain **moment of inertia and angular momentum** and apply their relationships to analyze the rotational motion of objects.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	please check student Teams
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Bring a charged devices, calculator, Physics notebook, ruler, pencil and coloured pens daily. Complete all graph and calculation work, participate actively, review feedback and upload required work to Teams.		

#الأسبوع Week # 4

Technology Integration دمج التكنولوجيا	Digital graphing or teacher-provided graph displays; calculators for gradients, areas and equations; motion videos or simulations where available; Teams for resources, feedback and submission.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: gradient, geometric area, algebra, rearranging equations, proportional reasoning and negative values. English: interpret motion descriptions and justify conclusions. UAE engineering and road safety: analyse transport data and stopping motion.
External Assessment Integration دمج التقييمات الخارجية	NGSS-style external assessment questions requiring students to analyse data, interpret multi-section graphs, calculate gradient and signed area, select kinematics equations, translate between representations and justify conclusions.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL Thinking: analyse patterns, connect representations and evaluate solutions. Communication: explain graph features and justify equation choices. Self-management: organise multi-step working and check signs and units. Collaboration: critique reasoning respectfully. Wellbeing: build confidence through scaffolded practice and purposeful challenge.
Standards المعايير	HS-PS2-1: Analyze data to support the claim that Newtons second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. Supporting alignment: students analyse motion data and graphical acceleration as the kinematic foundation for later force-motion modelling.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can the same journey be represented by equations and different motion graphs, and how can we verify that the representations are consistent?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1** ☐ **Must: Identify the horizontal and vertical components of a projectile's velocity.** ☐ **Should: Explain why horizontal velocity remains constant while vertical velocity changes due to gravitational acceleration.** ☐ **Could: Use kinematic equations to calculate displacement, velocity, and time for a horizontally launched projectile..** **Lesson 2:** ☐ **Must: Resolve the initial velocity into horizontal and vertical components using \(v_x=v\cos\theta\) and \(v_y=v\sin\theta\).** ☐ **Should: Calculate quantities such as time of flight, maximum height, horizontal range, and velocity.** ☐ **Could: Analyze multi-step projectile-motion problems and justify the choice of equations used.** **Lesson3:** ☐ **Must: Identify common forces, including weight, normal force, tension, friction, and applied force.** ☐ **Should: Draw and label accurate free-body diagrams and determine the resultant/net force.** ☐ **Could: Apply \(\Sigma F=ma\) to solve problems and explain how the net force affects an object's acceleration.** **Lesson 4 HL ONLY** ☐ **Must: Define moment of inertia and explain how the distribution of mass relative to the axis of rotation affects it.** ☐ **Should: Calculate angular momentum using \(L=I\omega\) and relate moment of inertia to rotational motion.** ☐ **Could: Apply conservation of angular momentum to explain and solve problems involving changes in moment of inertia and angular velocity.**
National Identity Domain	Belonging الإنتماء — Global Understanding العالمي التفاهم — History التاريخ

مجالات الربط بالهوية الوطنية	
	المجال**Area:** UAE Transportation, Road Safety, Innovation and Sustainable Infrastructure
	التطبيقImplementation: Students analyse Abu Dhabi transport motion using graphs, gradients, signed areas and kinematics equations, connecting accurate modelling to UAE road safety and smart mobility.